Homework 5

Atharva Chalke

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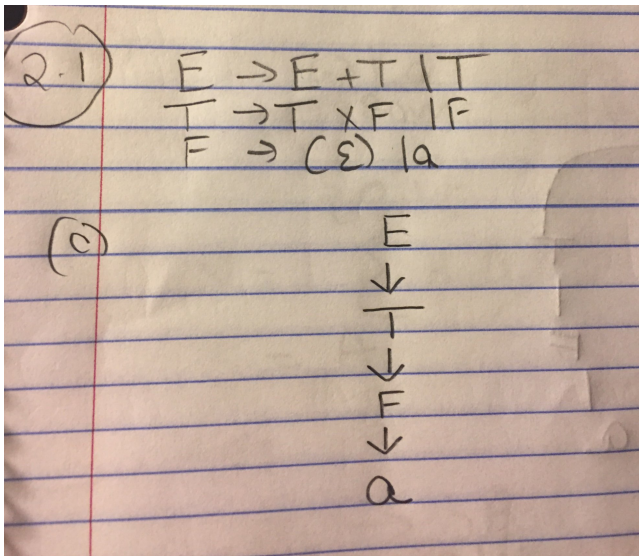
Problem 2.1a. *Proof.* **1 NFA**



2.1

$E \to E + T \mid T$

$T \to T \times F \mid F$

$F \to (E) \mid a$

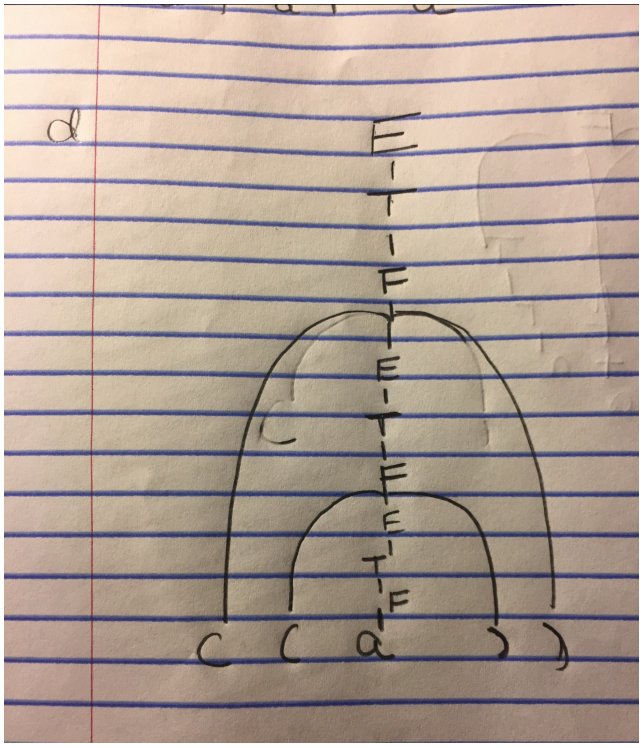
(c)

$E \downarrow T \downarrow F \downarrow a$

Scanned with CamScanner

□

Problem 2.1d. *Proof.* **2 NFA**



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Problem 2.4b. {w|w starts and ends with the same symbol}

Proof. $\Sigma = \{0,1\}$
V = {A,B}
Start state = A
Rules
A $\rightarrow$ 0B0|1B1|00|11
B $\rightarrow$ 0 | 1 | 0B | 1B

$\square$

Problem 3.4.

Proof. **string 00000**
Rejected
Steps - 6

string 000000
Rejected
Steps - 19 □